

A
SUMMER INTERNSHIP REPORT
ON
Software Development

Submitted by

Khadse Shraddha Shrikant . (236150316017)

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Athwagate, Surat-395001

CERTIFICATE

*This is to certify that the Summer Internship-1 report, submitted along with the project entitled **iOS Development** has been carried out by **Khadse Shraddha Shrikant (236150316017)** under my guidance in partial fulfilment for the degree of **Diploma in Information Technology 3rd Semester** of Gujarat Technological University, Ahmedabad during the academic year 2024-25. The student has successfully completed summer internship activity under my guidance.*

Mr. Vilas Vasava

Mentor, Lecturer
IT Department
GPG, Surat

Ms. Jigna J. Desai

Head
IT Department
GPG, Surat

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This report stands as a culmination of the knowledge, experiences, and efforts invested during the software Development internship.

Yours Sincerely,

Khadse shraddha Shrikant .(236150316017)

ABSTRACT

This report summarizes a long internship focused on software Development using python and c++. The internship aimed to provide practical experience in app development and enhance proficiency in python programming.

Throughout the internship, diverse projects were undertaken to explore various facets of software development, explores the integration of facets, a concept used to organize and visualize complex data,. Collaborative efforts and exposure to agile methodologies contributed to a deeper understanding of software development workflows.

The content begins with an introduction to Python fundamentals, including data structures, functions, and object-oriented programming principles. It then delves into the concept of facets, which allows developers to categorize and analyze different aspects or dimensions of data within their applications.

Through practical examples and case studies, this guide demonstrates how Python's libraries and frameworks can be utilized to implement facet-based software solutions effectively. Emphasis is placed on leveraging facets to enhance data organization, visualization, and user interaction within software projects.

The report details the projects undertaken, challenges faced, and lessons learned, offering insights into the practical application of python and C++ in creating functional software applications.

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Chapter-1: Introduction

1.1 Internship Summary:

- Concept Learned:

1. **Python Programming Language:** This included learning about syntax, data types, element, functions, control flow, and leveraging python features for efficient software development.
2. **App Design :** I learned the importance of user-centric design and its impact on the success of an application
3. **Software Development Tools :** integrated Development Environments (IDEs), Code Editors, and other development tools provided a comprehensive understanding of the software development system
4. **Debugging and Testing:** The internship provided exposure to debugging techniques, unit testing, and the importance of quality assurance in ensuring a seamless user experience.
5. **Version Control:** I learned to efficiently manage code repositories and collaborate effectively with team members.

1.2 Purpose:

- Concept Learned:

1. **Documentation of Learning and Experience:** It outlines the practical application of theoretical knowledge, the challenges faced, and the solutions devised.

2. **Demonstration of Skills:** It highlights specific projects worked on, the technologies used, and the achievements accomplished, serving as a testament to the intern's capabilities in Software development.
3. **Communication:** It serves as a means to communicate the experiences and insights gained during the internship to supervisors, mentors, or potential employers.
4. **Training Purposes:** The report can be used for Training purposes, benefiting other students, aspiring developers, or individuals seeking insights into software development.
5. **Professional Development and Career Growth:** It can be utilized during job interviews, demonstrating practical experience and expertise in Software development , thus enhancing career prospects.

1.3 Scope:

- Scope of iOS:

1. Programming Language Proficiency
2. Development Tools and Technologies
3. Software Development
4. internet of Things (IoT)
5. API Integration and Networking
6. Game Development
7. Debugging and Testing
8. Collaboration and Communication
9. Achievements and Future Goals
10. Web Development

Chapter-2: Details of Technology / Platform

2.1 python Introduction:

→ python is a high-level, interpreted programming language known for its simplicity, readability, and versatility.

→ Created by Guido van Rossum and first released in 1991, Python's design philosophy emphasizes code readability and ease of use, making it an excellent choice for both beginners and experienced programmers..

→ Python's straightforward syntax resembles plain English, making it easy to read and write. This simplicity reduces the learning curve for new programmers and allows for rapid development.

→ Python supports object-oriented programming (OOP) paradigms, which allows for the creation of reusable code and the development of complex applications.

→ Python comes with a vast standard library that includes modules and packages for a wide range of tasks, from file I/O to web development, data manipulation, and more.

- → Python emphasizes code readability with its clear syntax and indentation. Uses English keywords frequently where other languages use punctuation. Python code is executed line by line, making it easy to test and debug. Variable types are determined at runtime, allowing for more flexibility. Python comes with a vast library of modules and packages, supporting many common tasks. A large and active community provides extensive documentation, third-party packages, and tutorials.

Versions of python:

- Python has several versions, each introducing new features, improvements, and sometimes breaking changes. Here are the major versions of Python:
- **Python 1.x:** The initial version released in 1991.
- **Python 2.x:** Released in 2000, Python 2.x introduced many new features and was the primary version used until Python 3.x. The final release was Python 2.7, which was supported until January 1, 2020.
- **Python 3.x:** Released in 2008 to address and fix fundamental design flaws in the language. This version is not backward-compatible with Python 2.x. Python 3.0 The initial release. Python 3.1 - 3.3 Introduced various enhancements and performance improvements. Python 3.4 Introduced the `asyncio` module. Python 3.5 Introduced `async` and `await` keywords.
- Python 3.6 Introduced formatted string literals (f-strings). Python 3.7 Improved data classes, context variables, and other features. Python 3.8 Introduced the walrus operator (`:=`). Python 3.9 Added type hinting improvements, new syntax features.
- python 3.10 (October 2021) Introduced pattern matching (similar to switch/case statements in other languages) and more precise error messages. Python 3.11 (October 2022) Focused on performance improvements, potentially doubling the speed of certain operations.
- **input() Function:** In Python 2, `input()` evaluates input as code. Use `raw_input()` for string input. In Python 3, `input()` always returns a string.

Example-

Code:-

```
print("Hello, World!")
```

Output:-

Hello, World!

- **Basic python structure :**

- Python is a high-level, interpreted programming language known for its simplicity and readability. It supports multiple programming paradigms, including procedural, object-oriented, and functional programming.
- Readable and Maintainable Code: Python's clean syntax makes it easy to read and write code.
- Dynamic Typing: Variables in Python are not declared with a specific type. The type is determined at runtime.
- Interpreted Language: Python code is executed line by line, which makes debugging easier. Content The information (text or other elements) contained between the opening and closing tags of an HTML element.
- Extensive Standard Library: Python comes with a vast standard library that supports many common programming tasks.

- **Our first python code :**

```
x = 5
y = 3.14
name = "Alice"
is_active = True
print(x)
print(y)
print(name)
print(is_active)
```

Explanation:

Python program structure :-

Comments:

Comments are crucial for documenting your code. They provide explanations, notes, and help others (or yourself in the future) understand the code's logic. Multi-line comments or docstrings Enclosed in triple quotes (""" or '''). Docstrings are used to describe modules, functions, classes, or methods.

```
# Single-line comment

"""
Multi-line comment
or docstring
"""
```

Imports :-

Imports bring functionality from external modules or packages into your program. They should be placed at the top of the file, right after any module-level docstring or comments. Absolute imports Directly specify the module's path from the project's root. Relative imports Import modules relative to the current module.

```
import math
from datetime import datetime
```

Variables and data types :-

Variables hold data that can be modified during execution. Constants are variables that should not be changed once defined. Naming conventions: Use descriptive names. Constants are typically named in all caps. Variables store data and can change value throughout the program. Python supports various data types like integers, floats, strings, and Booleans.

```
# Variable assignment
x = 5          # Integer
y = 3.14       # Float
name = "Alice" # String
is_active = True # Boolean
```

Data structures :-

Python provides built-in data structures like lists, tuples, sets, and dictionaries to store collections of data. Lists: Ordered collection, mutable. Tuples: Ordered collection, immutable. Sets: Unordered collection of unique elements. Dictionaries: Key-value pairs.

```
# List
fruits = ["apple", "banana", "cherry"]

# Tuple
coordinates = (10.0, 20.0)

# Set
unique_numbers = {1, 2, 3, 4}

# Dictionary
student = {"name": "Alice", "age": 20}
```

Conditional statement :-

Conditional statements allow you to execute different code based on certain conditions

```
age = 18
if age >= 18:
    print("You are an adult.")
else:
    print("You are a minor.")
```

Chapter-3: Project Details

Project Title: Create Multiple Demo

- ❖ Sign In – Sing Up in create software
- ❖ Picker View demo in python
- ❖ Table View demo in python
- ❖ Collection View demo in python

Technologies Used:

1. Used python programming language
2. Xcode IDEL code
3. Third-party libraries

Chapter-4: Implementation

❖ python Introduction.

A Hello world is a simple program that prints Hello,world! On the Screen.This Program is used to introduce a new programming language to beginners.

❖ Built in Data Types:-int , float , double , bool , var , val

Int :- To Store Positive or Negative Number including zero

Float :- To Store Decimal or Fractional Number

Double :-To Store Decimal or Fractional Number

Bool :- TO Store Only True or False Values

Var :- Use to Declare Variable

Val :- Use to Declare Constants

❖ Arithmetic Operators

+ :- Addition ,

- :- Subtraction ,

*** :-** Multiplication ,

/ :- Division ,

// :- Integer division,

% :- Modulus,

**** :-** Exponentiation.

➤ Assignment Operators

= :- Assignment Operator

➤ Remainder Operator

% :- Modulo ,

❖ Unary Operator

(-) :- Unary Minus Operator

(+) :- Unary Plus Operator

❖ Compound Assignment Operator

+= :- Addition and assignment operator

-= :- Subtraction and assignment operator

*= :- Multiplication and assignment operator

/= :- Division and assignment operator

//= :- Integer division and assignment operators

%= :- Modulus and assignment operators

**= :- Exponentiation and assignment operators

&= :- Bitwise AND and assignment operators

❖ Comparison Operators

`== :-` Equal to

`!= :-` Not Equal to

`> :-` Greater than

`< :-` Less than

`>= :-` Greater than or equal to

`<= :-` Less than or equal to

❖ Logical Operators

`and :-` Logical AND

`or :-` Logical OR

`not :-` Logical NOT

❖ Bitwise Operators

`& :-` Bitwise AND

`| :-` Bitwise OR

`^ :-` Bitwise XOR

`~ :-` Bitwise NOT

`<< :-` Left shift

`>> :-` Right shift

❖ If Statement

if expression:

`//statement`

```
temperature = 75

# Check the value of the variable using
if temperature > 70:
    print("It's a warm day.")
else:
    print("It's a cool day.")
```

Output:-

```
It's a warm day.
```

❖ If – else Statement

if expression:

 //statement

else :

 //statement

```
x = 3
if x > 5:
    print("x is greater than 5")
else:
    print("x is not greater than 5")
```

Output:-

```
x is greater than 5
```

❖ If – else- If Statement

if expression1:

 //statement

else if expression2:

```
//statement }  
else :  
    //statement
```

```
# Example of an if-elif-else statement  
x = 7  
  
if x > 10:  
    print("x is greater than 10")  
elif x > 5:  
    print("x is greater than 5 but less than  
else:  
    print("x is 5 or less")
```

Output:- x is greater than 5 but less than or equal to 10

❖ Nested If Statement

```
if expression1:  
    //Statements if  
    expression2:  
        // Statements
```

```
# Example of nested if statements
x = 8

if x > 5:
    print("x is greater than 5")
    if x > 10:
        print("x is also greater than 10")
    else:
        print("x is not greater than 10")
else:
    print("x is 5 or less")
```

Output:- x is greater than 5
 x is not greater than 10

❖ Nested If – Else Statement

```
if expression1:
    if expression2:
        // Statements
    else:
        //Statements
else:
    //Statements
```

```
# Example of nested if-else statements
x = 12

if x > 10:
    print("x is greater than 10")
    if x > 15:
        print("x is also greater than 15")
    else:
        print("x is 10 to 15")
else:
    if x > 5:
        print("x is greater than 5 but 10 or less")
    else:
        print("x is 5 or less")
```

Output:-

```
x is greater than 10
x is 10 to 15
```

❖ Break Statement

break : It is used to terminate the statement and comes out from the loop

```
# Example of using a break statement in a loop
for i in range(10):
    if i == 5:
        print("Breaking the loop")
        break
    print(i)
```

Output:-

```
0
1
2
3|
4
Breaking the loop
```

❖ Continue Statement

Continue: It is used to skip the statement and continue the loop

```
# Example of using a continue statement in a loop
for i in range(1, 6):
    if i == 3:
        print("Skipping", i)
        continue
    print("Current number:", i)
```

Output:-

```
Current number: 1
Current number: 2
Skipping 3
Current number: 4
Current number: 5
```

❖ For In Loop

for val in val_name :

//Statements

```
# Example of a for loop
fruits = ["apple", "banana", "cherry"]

for fruit in fruits:
    print(fruit)
```

Output:-

```
apple  
banana  
cherry
```

❖ While Loop

while (Test Expression) :

 // Statements

```
# Example of a while loop  
i = 1  
  
while i <= 5:  
    print(i)  
    i += 1
```

Output:-

```
1  
2  
3  
4  
5
```

❖ Function

❖ Function with Multiple Arguments


```
def add(a, b):  
    """  
    This function returns the sum of two numbers.  
    """  
    return a + b  
  
# Call the function  
result = add(3, 5)  
print(f"The sum is: {result}")
```

Output:-

The sum is: 8

❖ Function with Variable Number of Arguments

```
def greet(*names):  
    """  
    This function greets all persons whose names  
    """  
    for name in names:  
        print(f"Hello, {name}!")  
  
# Call the function with multiple arguments  
greet("Alice", "Bob", "Charlie")
```

Output:-

Hello, Alice!

Hello, Bob!

Hello, Charlie!

❖ Lambda Functions

```
# Define a lambda function
add = lambda a, b: a + b

# Call the lambda function
result = add(3, 5)
print(f"The sum is: {result}")
```

Output:-

The sum is: 8

❖ Array:

❖ Creating a NumPy Array

```
import numpy as np

# Creating a NumPy array
arr = np.array([1, 2, 3, 4, 5])
print("NumPy array:", arr)
```

Output:-

NumPy array: [1 2 3 4 5]

❖ Class :

To define a class in Python, you use the `class` keyword followed by the class name and a colon. Inside the class, you define methods and attributes.

```
class Person:
    # Constructor method
    def __init__(self, name, age):
        self.name = name # Instance variable
        self.age = age # Instance variable

    # Method to display person's details
    def display(self):
        print(f"Name: {self.name}, Age: {self.age}")

# Creating an instance of the Person class
person1 = Person("Alice", 30)

# Accessing methods and attributes
person1.display()
print(f"Name: {person1.name}, Age: {person1.age}")
```

Output:-

Name: Alice, Age: 30

Name: Alice, Age: 30

❖ Class and Instance Variables

```
class Person:
    species = "Homo sapiens" # Class variable

    def __init__(self, name, age):
        self.name = name # Instance variable
        self.age = age # Instance variable

    def display(self):
        print(f"Name: {self.name}, Age: {self.age}, Species: {Person.species}")

# Creating instances
person1 = Person("Alice", 30)
person2 = Person("Bob", 25)

# Accessing class and instance variables
person1.display()
person2.display()
```

Output:-

```
Name: Alice, Age: 30, Species: Homo sapiens
Name: Bob, Age: 25, Species: Homo sapiens
```

❖ Design in IDLE Xcode as Follow :

- Basic Calculator **Screen**
- Basic calculator implement in calculator demo

```
Simple Calculator
Select operation:
1. Add
2. Subtract
3. Multiply
4. Divide
Enter choice (1/2/3/4): 1
Enter first number: 2
Enter second number: 65
2.0 + 65.0 = 67.0
Do you want to perform another calculation? (yes/no): yes
Enter choice (1/2/3/4): 4
Enter first number: 50
Enter second number: 10
50.0 / 10.0 = 5.0
Do you want to perform another calculation? (yes/no): yes
Enter choice (1/2/3/4): 2
Enter first number: 2
Enter second number: 3
2.0 - 3.0 = -1.0
Do you want to perform another calculation? (yes/no): no
```

[Figure1: Basic calculator screen]

❖ Design in IDEL Xcode as Follow :

- Create Word guess game in screen demo
- Enter random number and guess word

```
What is your name? Gautam
Good Luck!  Gautam
Guess the characters
—
—
—
—
—
guess a character:g
g
—
—
—
—
—
guess a character:e
g
e
e
—
—
—
guess a character:k
g
e
e
k
—
—
guess a character:s
g
e
e
k
s
You Win

The word is:  geeks
```

[Figure2: Word guess game screen]

❖ Design in IDEL Xcode as Follow :

- Create Form of college in screen demo
- Create the form of student enter the college

```
College Application Form
-----
Enter your full name: khadse shraddha shrikant
Enter your age: 18
Enter your address: 482,sonal socitey udhana,surat
Enter your phone number: 7048168003
Enter your email address: skhadse111@gmail.com
Enter your high school name: suman hight school no.06
Enter your GPA: 84
Enter your intended major: information technology

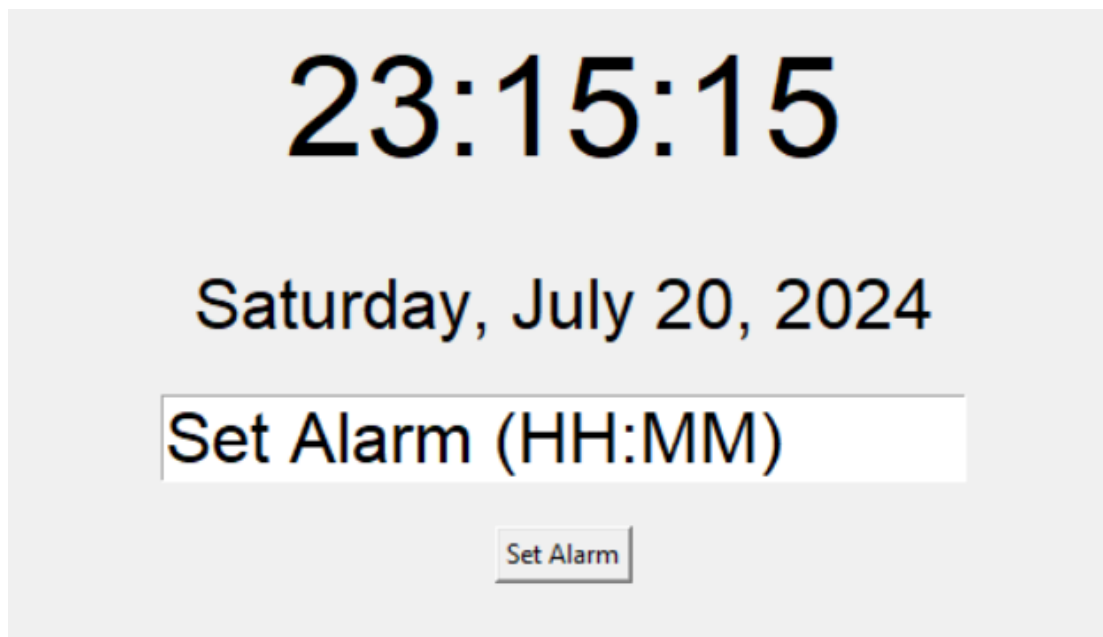
Thank you for your application! Here is the information you provided:
Name: khadse shraddha shrikant
Age: 18
Address: 482,sonal socitey udhana,surat
Phone Number: 7048168003
Email: skhadse111@gmail.com
High School: suman hight school no.06
GPA: 84
Intended Major: information technology
```

Full Name	dse shraddha shrikant
Age	18
Address	I socitey udhana,surat
Phone Number	704868003
Email Address	skhadse111@gamil.cor
High School Name	an hight school no. 06
GPA	84
Intended Major	formation technology
Submit	

[Figure3: Form of college screen]

❖ Design in IDEL Xcode as Follow :

- Create smart watch in screen demo
- Display the time of smart way



[Figure 4: smart watch screen]

❖ Design in IDEL Xcode as Follow :

- Create student information table type in screen demo

ID	Name	Age	Grade
1	Komal patil	20	A
2	Ayush bonde	22	B
3	Harshali patil	21	A
4	David Wilson	23	C

[Figure 5: student information table type screen]

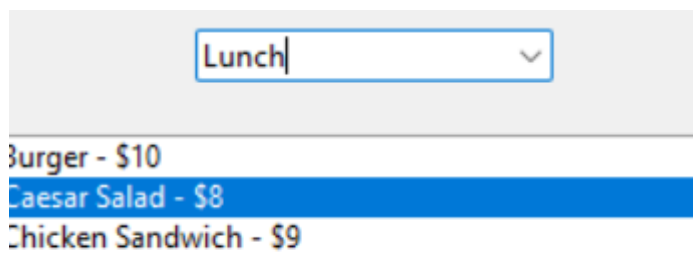
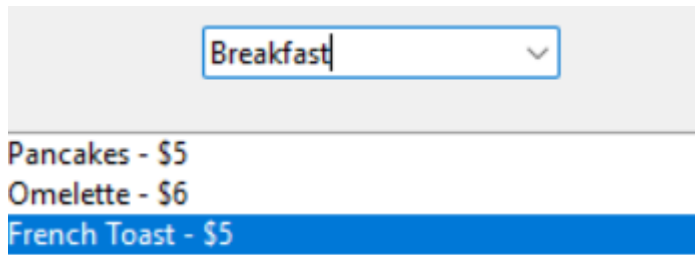
❖ Design in IDEL Xcode as Follow :

- Create Hotel menu list screen demo

```
Hotel Menu
-----
1. Breakfast
2. Lunch
3. Dinner
4. Beverages
5. Exit
Select an option (1-5): 4

Menu Items:
Coffee - $3
Tea - $2
Soft Drink - $2

Hotel Menu
-----
1. Breakfast
2. Lunch
3. Dinner
4. Beverages
5. Exit
Select an option (1-5): 5
Exiting...
```



[Figure 6: Hotel menu list screen]

❖ Design in IDEI Xcode as Follow:

- Create the Tic-Tac-Toe game

```
[[0 0 0]
 [0 0 0]
 [0 0 0]]
Board after 1 move
[[0 0 0]
 [1 0 0]
 [0 0 0]]
Board after 2 move
[[0 0 2]
 [1 0 0]
 [0 0 0]]
Board after 3 move
[[1 0 2]
 [1 0 0]
 [0 0 0]]
Board after 4 move
[[1 0 2]
 [1 0 0]
 [2 0 0]]
Board after 5 move
[[1 0 2]
 [1 0 1]
 [2 0 0]]
Board after 6 move
[[1 0 2]
 [1 2 1]
 [2 0 0]]
Winner is: 2
```

[Figure 7: Create Tic-Tac-Toe game]

Chapter-5: Conclusion

This internship has been an excellent and rewarding experience. I can conclude that there have been a lot I've learnt from my work at the training and research centre.

Needless to say, the technical aspects of the work I've done are not flawless and could be improved provided enough time.

As someone with no prior experience in python programming whatsoever I believe my time spent in training and discovering new languages was well worth it and contributed to finding an acceptable solution to an important aspect of software design and development.

Two main things that I've learned the importance of time management skills and self motivation. Working with web development language has increased my interest in them, hence prompting me to transfer to the software Design and Development course at my college

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